Jordan McCarney

Curriculum Vitae

School of Mathematical Sciences,
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Education

2026 **Ph.D. in Mathematics**, University College Cork (UCC).
Thesis: *Mathematical aspects of internal wave-current interactions.*

2022 **B.Sc. in Theoretical Physics**, University of Galway (UG).

Employment

2026/27 **Postdoctoral Researcher**, University College Cork.

Research Interests

My research interests concern the nonlinear analysis of ordinary and partial differential equations, particularly those that arise in the context of geophysical fluid flows in the ocean and atmosphere. Much of my work addresses free-boundary problems. In general though, my interests lie in the use of rigorous mathematics to answer questions which arise in applications.

Publications

2026 [1] D. Devine and **J.McCarney**. Flow underlying coupled surface and internal waves with rotational lower layer. Submitted.

[2] **J.McCarney** and T. Lyons. Nonlinear temperature dependent enthalpies in exact nonlinear mountain waves, *Differential and Integral Equations*, 39(7/8).

2025 [3] T.Lyons and **J.McCarney**. Temperature dependent precipitation in exact nonlinear mountain waves, *Journal of Mathematical Fluid Mechanics*, 27(3).

2024 [4] **J.McCarney.** Nonhydrostatic internal waves in the presence of mean currents and rotation, *Journal of Mathematical Physics*, 65(4).

2023 [5] **J.McCarney.** Exact internal waves in the presence of mean currents and rotation, *Journal of Mathematical Physics*, 64(7).

Awards and Prizes

2026 **SIAM Travel Award** - approx. USD 1,000
Competitive grant awarded by Society for Industrial and Applied Mathematics.

2024 **Postgraduate Publication of the Year**
UCC College of Science, Engineering and Food Science.
I received this award for my independent research article [4].

Invited Research Presentations

I have been invited to deliver seminars in ten universities or research insitutes in eight countries:

2026 19 Nov — **Partial Differential Equations and Mathematical Physics Seminar** University of Vienna, Austria.

26-28 Aug — **Wave-NED Conference on Mathematics of Wave Phenomena** Delft University of Technology, Delft, The Netherlands.

26-29 May — **SIAM Conference on Nonlinear Waves and Coherent Structures** Concordia University, Montréal, Québec, Canada.

11 Feb — **Mathematics and Statistics Seminar** Technological University Dublin, Ireland.

2025 18 Nov — **Oberseminar Analysis und Theoretische Physik** Leibniz Universität Hannover, Germany.

16-19 Jun — **Modelling of Fluid Propagation: Math. theory and numerical approximation** International Centre for Mathematical meetings, Castro Urdiales, Spain.

2 Apr — **Mathematics and Physics Seminar** South East Technological University, Waterford, Ireland.

26 Mar — **Partial Differential Equations Seminar** University College Cork, Ireland.

2024 28-30 Aug — **Water Waves-Mathematical Theory and Applications** University of Plymouth, UK.

27-31 May — **EWM-EMS Summer School: Water Waves and Nonlinear Dispersive Equations** Mittag-Leffler Institute, Stockholm, Sweden.

Invited Visits to Research Institutes

2025 16-19 Jun — **International Centre for Mathematical meetings**, Castro Urdiales, Spain. "Modelling of fluid propagation: mathematical theory and numerical approximation".

2024 27-31 May — **Mittag-Leffler Institute**, Djursholm, Sweden. EWM-EMS Summer School: "Water Waves and Nonlinear Dispersive Equations".

2023 22-25 May — **Erwin Schrödinger Institute**, Vienna, Austria. "The Dynamics of Planetary-scale Fluid Flows".

Supervision

2025/26 — **B.Sc. Mathematical Sciences Thesis** - Ms. Megan Flynn, Thesis: *Topological degree theory and its applications.*

Teaching

UCC **Teaching Assistant - Delivered lectures and all tutorial classes:**
MA4052 Functional Analysis
AM4063 Partial Differential Equations with Applications II
AM3063 Partial Differential Equations with Applications I
AM3052 Introduction to Fluid Mechanics and Wave Theories
AM2071 Transform and Variational Methods

UCC **Tutor - Delivered all tutorial classes:**
AM2061 Computer Modelling and Numerical Techniques
AM1052 Introduction to Mechanics

UG **Tutor - Drop in problem-solving sessions for 1st year students:**
Physics Undergraduate Problem Solving.

UCC **Pedagogical Development**
I successfully completed "PG6003 Teaching and Learning for Graduate Studies".

Referee Activity

I have been invited to act as a referee for the following international research journals:

- *Journal of Differential Equations*,
- *Journal of Mathematical Physics*,
- *Nonlinear Analysis: Real World Applications*,
- *Qualitative Theory of Dynamical Systems*,
- *Pure and Applied Geophysics*,
- *European Journal of Mechanics / B Fluids*.